 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1 COMP Reflex arc components

You should be able to: Diagram a reflex arc naming the receptor, sensory neuron, integrating center, motor neuron, and effector, and classify the reflex as monosynaptic or polysynaptic.

PROMPT ☐ A ☐ B PASS 1 COLOUR ______ PASS 2 COLOUR ______ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

2 COMP Muscle spindle and stretch reflex

You should be able to: Explain how the muscle spindle detects stretch and trace the stretch reflex including reciprocal inhibition of the antagonist.

PROMPT ☐ A ☐ B PASS 1 COLOUR ______ PASS 2 COLOUR ______ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

3 Golgi tendon organ

COMP

You should be able to: Contrast the stimulus and the reflex response of the Golgi tendon organ with those of the muscle spindle.

PROMPT ☐ A ☐ B PASS 1 COLOUR ____________ PASS 2 COLOUR ____________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

4 Withdrawal and crossed extensor reflexes

COMP

You should be able to: Trace the flexor withdrawal reflex and the crossed extensor reflex and explain how the two together preserve balance.

PROMPT ☐ A ☐ B PASS 1 COLOUR ____________ PASS 2 COLOUR ____________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

5

COMP

Ascending and descending pathways

You should be able to: Compare the dorsal column, spinothalamic, and corticospinal pathways by the information carried, the side of decussation, and the deficit produced by a lesion.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

6

COMP

Cerebrospinal fluid and the blood brain barrier

You should be able to: State the functions of cerebrospinal fluid and explain how the blood brain barrier selects what reaches neurons, including why lipid soluble drugs cross readily.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

7 COMP — Reflex testing and reaction time

You should be able to: Elicit and grade deep tendon reflexes and measure reaction time to distinguish a reflex response from a voluntary response by latency.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

8 COMP — Sensory transduction

You should be able to: Explain how a stimulus is converted into a receptor potential and how that graded potential becomes a train of action potentials in the sensory neuron.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ______________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

9

COMP

Receptor classification

You should be able to: Classify a receptor by stimulus type as a mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, or nociceptor and give a body location for each.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

10

COMP

Stimulus coding

You should be able to: Explain how the nervous system encodes modality, location, intensity, and duration and identify the labeled line principle.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

11 Receptive fields and acuity

COMP

You should be able to: Relate receptive field size and lateral inhibition to two point discrimination and predict which body regions have the finest acuity.

PROMPT ☐ A ☐ B PASS 1 COLOUR ______ PASS 2 COLOUR ______ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

12 Receptor adaptation

COMP

You should be able to: Distinguish tonic from phasic receptors by their adaptation rate and match each to a stimulus the body needs to monitor continuously or only at onset.

PROMPT ☐ A ☐ B PASS 1 COLOUR ______ PASS 2 COLOUR ______ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

NAME ____________________ DATE ______________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

13 COMP — Somatosensory pathways

You should be able to: Trace touch and pain information from receptor to cortex and explain why the somatosensory homunculus is disproportionate.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

14 COMP — Pain and its modulation

You should be able to: Distinguish fast and slow pain by fiber type, explain referred pain by convergence, and describe how gate control and endogenous opioids reduce pain transmission.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

15 COMP

Tactile mapping and adaptation testing

You should be able to: Measure two point discrimination at several body sites and show receptor adaptation and referred sensation experimentally.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

16 COMP

Optics and accommodation

You should be able to: Explain refraction by the cornea and lens and describe how accommodation and the pupillary reflex focus near and far images.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

17 Phototransduction

COMP

You should be able to: Trace phototransduction from photon absorption by retinal through the change in cyclic GMP to the decrease in glutamate release and explain why photoreceptors hyperpolarize to light.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

18 Rods cones and visual processing

COMP

You should be able to: Compare rods and cones by sensitivity and acuity and trace the visual pathway to predict the field deficit produced by a lesion at the optic nerve or the optic chiasm.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

19 Refractive errors and clinical vision

COMP

You should be able to: Explain myopia, hyperopia, presbyopia, and astigmatism in terms of focal point and eyeball or lens shape and state the corrective lens for each.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

20 Auditory transduction

COMP

You should be able to: Trace sound from the tympanic membrane through the ossicles and cochlear fluid to hair cell bending and auditory nerve firing and explain how pitch and loudness are coded.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

21 COMP — Conductive and sensorineural hearing loss

You should be able to: Distinguish conductive from sensorineural hearing loss by the site of the lesion and by the expected Weber and Rinne results.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

22 COMP — Equilibrium

You should be able to: Explain how the semicircular canals detect angular acceleration and how the utricle and saccule detect linear acceleration and head position.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

23 COMP — Taste and smell

You should be able to: Compare gustatory and olfactory transduction and explain why olfactory input reaches the limbic system without a thalamic relay.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

24 COMP — Vision testing

You should be able to: Perform visual acuity, blind spot, accommodation, and color vision testing and interpret each result.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

NAME ____________________ DATE ______________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

25 COMP — Hearing and equilibrium testing

You should be able to: Perform Weber and Rinne tuning fork tests and a balance test and interpret the findings.

PROMPT ☐ A ☐ B PASS 1 COLOUR __________ PASS 2 COLOUR __________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

26 COMP — Motor control hierarchy

You should be able to: Order the levels of motor control from spinal reflex through brainstem to cortex and state what each level contributes to a voluntary movement.

PROMPT ☐ A ☐ B PASS 1 COLOUR __________ PASS 2 COLOUR __________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ______________ DATE __________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

27 Corticospinal pathway

COMP

You should be able to: Trace the corticospinal pathway from the primary motor cortex to the skeletal muscle and identify where it crosses.

PROMPT ☐ A ☐ B PASS 1 COLOUR ______ PASS 2 COLOUR ______

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

28 Cerebellum and basal ganglia

COMP

You should be able to: Contrast the contributions of the cerebellum and the basal ganglia to movement and match a described motor sign to the structure involved.

PROMPT ☐ A ☐ B PASS 1 COLOUR ______ PASS 2 COLOUR ______

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

29 COMP Upper and lower motor neuron signs

You should be able to: Distinguish upper from lower motor neuron lesions by tone, reflexes, and muscle bulk and assign a set of findings to the correct level.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

30 COMP Autonomic and somatic organization

You should be able to: Contrast the somatic motor system with the autonomic two neuron pathway by neuron count and by effector tissue.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

31 COMP Sympathetic and parasympathetic divisions

You should be able to: Compare the two autonomic divisions by CNS origin and ganglion location and predict the effect of each on heart rate and on the gastrointestinal tract.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

32 COMP Autonomic neurotransmitters and receptors

You should be able to: Match nicotinic, muscarinic, alpha, and beta receptors to their neurotransmitter, their location, and the response each produces.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

 NAME ____________ DATE ____________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

33 Dual innervation and tonic control

COMP

You should be able to: Explain dual innervation and autonomic tone and predict the effect of removing one division from a dually innervated organ.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

34 Autonomic pharmacology

COMP

You should be able to: Predict the effect of a beta blocker or a muscarinic antagonist on a named organ from the receptor it targets.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

NAME ____________________ DATE ______________

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

35

COMP

Autonomic function testing

You should be able to: Measure heart rate variability and the response to a cold pressor or orthostatic challenge and interpret the result as sympathetic or parasympathetic dominance.

PROMPT ☐ A ☐ B PASS 1 COLOUR __________ PASS 2 COLOUR __________ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops